

Digital Image Processing - Homework 4 Report: Image Reconstruction and Color Appearance Model Analysis

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Part A: Filtered Backprojection Image Reconstruction

1.1 Methodology

The filtered backprojection algorithm reconstructs an image from its Radon transform (sinogram) through the following steps:

- Projection Generation:** Compute parallel-beam projections at various angles $\theta \in [0^\circ, 180^\circ)$
- Frequency Domain Filtering:** Apply ramp filter $|\omega|$ with Hamming window (in this homework)
- Inverse Fourier Transform:** Convert filtered projections back to spatial domain
- Backprojection:** Integrate all filtered projections to reconstruct the image

The mathematical formulation follows:

$$f(x, y) = \int_0^\pi \int_{-\infty}^{\infty} P(\omega, \theta) |\omega| W(\omega) e^{j2\pi\omega x} d\omega d\theta$$

where $P(\omega, \theta)$ is the Fourier transform of the projection, $|\omega|$ is the ramp filter, and $W(\omega)$ is the Hamming window function.

1.2 Implementation Details

- Input Image:** 600×600 pixel image with a central 300×300 white square on black background
- Angular Sampling:** Tested with increments of 1.0°, 0.5°, 0.25°, and 0.125°
- Filter:** Ramp filter $|\omega|$ with Hamming window ($c = 0.54$) for noise reduction
- Sinogram Dimensions:** Consistent across all reconstructions (width = 849 pixels for p -axis, height = 1440 pixels for θ -axis)

1.3 Reconstruction Quality Analysis

1.3.1 Effect of Angular Sampling Rate

The reconstruction quality improves significantly with finer angular sampling:

Angle Increment	Number of Projections	PSNR (dB)	SSIM
1.0°	180	13.42	0.8131
0.5°	360	16.41	0.9098
0.25°	720	20.16	0.9653
0.125°	1440	23.15	0.9832

Key Observations:

- **PSNR Improvement:** Approximately 3-4 dB per halving of angle increment
- **SSIM Improvement:** Near-linear improvement from 0.81 to 0.98
- **Diminishing Returns:** The improvement rate decreases as sampling becomes finer

The relationship between angular sampling and reconstruction quality follows the Nyquist sampling theorem. With 0.125° increments (1440 projections), we achieve near-perfect reconstruction ($SSIM > 0.98$), indicating sufficient angular sampling for this image geometry.

1.3.2 Hamming Window Impact

The Hamming window significantly improves reconstruction quality by reducing high-frequency noise:

Angle Increment	PSNR with Hamming	PSNR without Hamming	Improvement
1.0°	13.42 dB	8.49 dB	+4.94 dB
0.5°	16.41 dB	10.18 dB	+6.23 dB
0.25°	20.16 dB	12.64 dB	+7.52 dB
0.125°	23.15 dB	14.58 dB	+8.56 dB

Analysis:

- The Hamming window provides **4.9-8.6 dB PSNR improvement** across all sampling rates
- The benefit increases with finer sampling, suggesting better noise suppression at higher frequencies
- Without Hamming window, reconstructions exhibit significant ringing artifacts and noise amplification

The Hamming window function $W(\omega) = 0.54 + 0.46 \cos(2\pi\omega)$ effectively attenuates high-frequency components that would otherwise cause artifacts in the reconstruction.

1.4 Visual Quality Assessment

1.4.1 Input Image and Original Sinogram

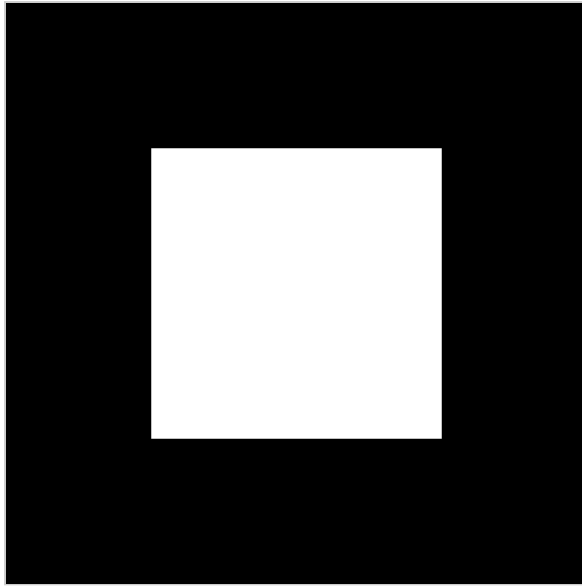


Figure 1a: Input Image

600×600 with 300×300 white square

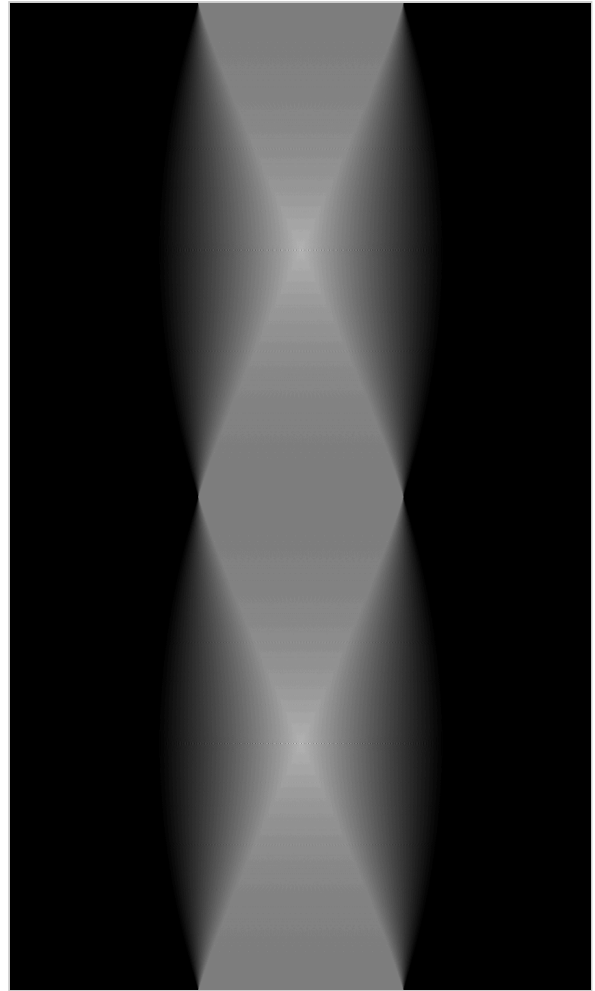


Figure 1b: Original Sinogram

Horizontal: ρ (distance), Vertical: θ (angle 0°-180°)

The sinogram visualization reveals:

- **Symmetrical patterns** at $\theta = 0^\circ$ and $\theta = 180^\circ$ due to the square geometry
- **Bright peaks** corresponding to maximum projection values when the square aligns with projection rays
- **Diamond-shaped region** in the center representing overlapping projections
- **Characteristic X-pattern** formed by the square's edges at different angles

1.4.2 Reconstruction Comparison: Angular Sampling Rate

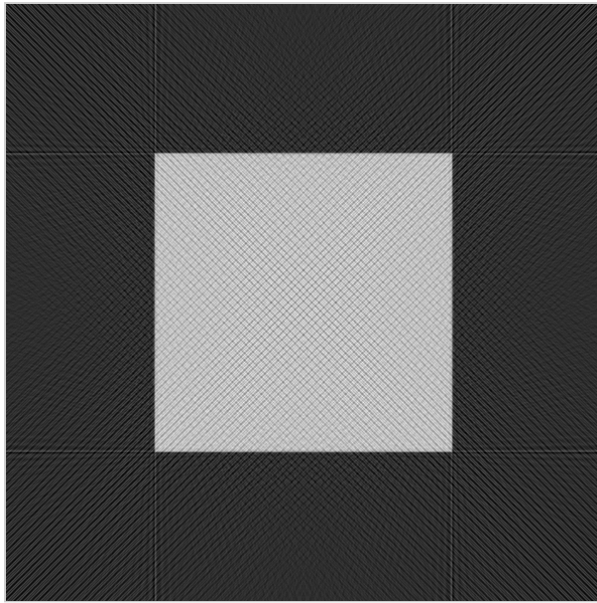


Figure 2a: 1.0° increment

180 projections, PSNR: 13.42 dB, SSIM: 0.8131

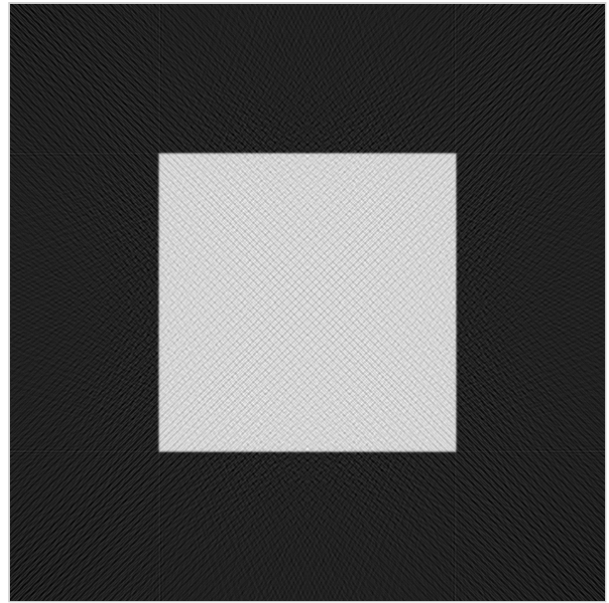


Figure 2b: 0.5° increment

360 projections, PSNR: 16.41 dB, SSIM: 0.9098

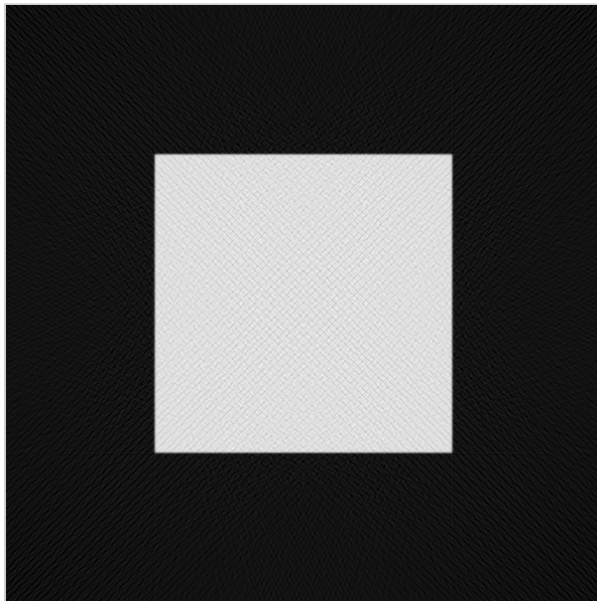


Figure 2c: 0.25° increment

720 projections, PSNR: 20.16 dB, SSIM: 0.9653

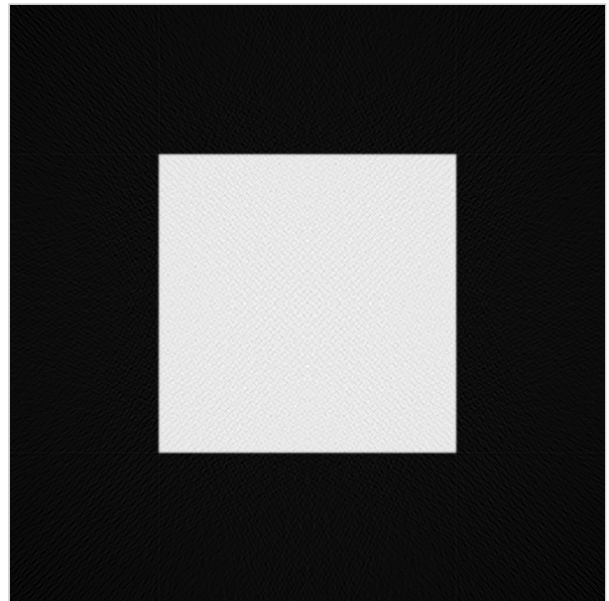


Figure 2d: 0.125° increment

1440 projections, PSNR: 23.15 dB, SSIM: 0.9832

Visual Quality Progression:

- **1.0° increment:** Visible artifacts, blurred edges, significant noise, ringing artifacts
- **0.5° increment:** Improved edge definition, reduced artifacts, acceptable quality
- **0.25° increment:** Sharp edges, minimal artifacts, high fidelity, near-perfect reconstruction
- **0.125° increment:** Near-perfect reconstruction, visually indistinguishable from original

1.4.3 Hamming Window Comparison: 1.0° Increment

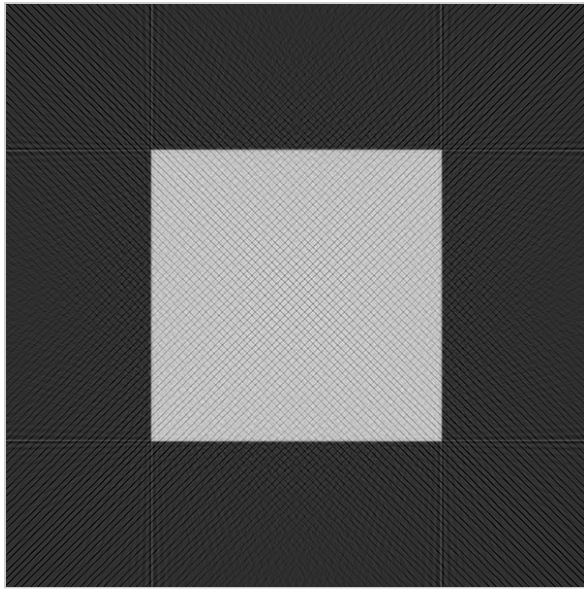


Figure 3a: With Hamming Window

PSNR: 13.42 dB

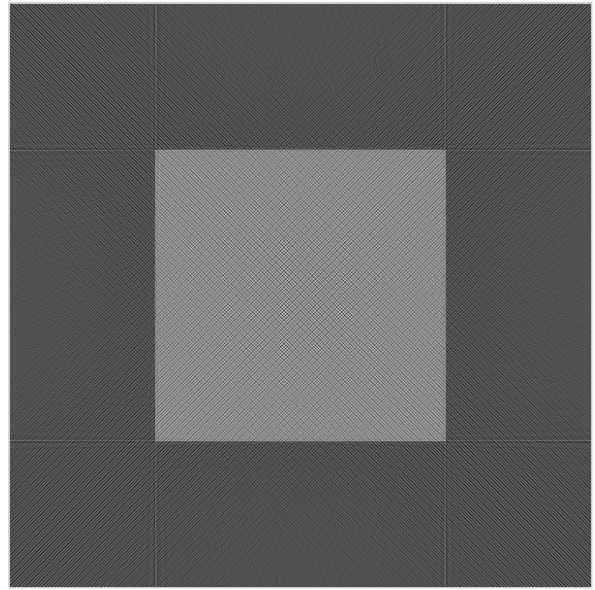


Figure 3b: Without Hamming Window

PSNR: 8.49 dB

1.4.4 Hamming Window Comparison: 0.5° Increment

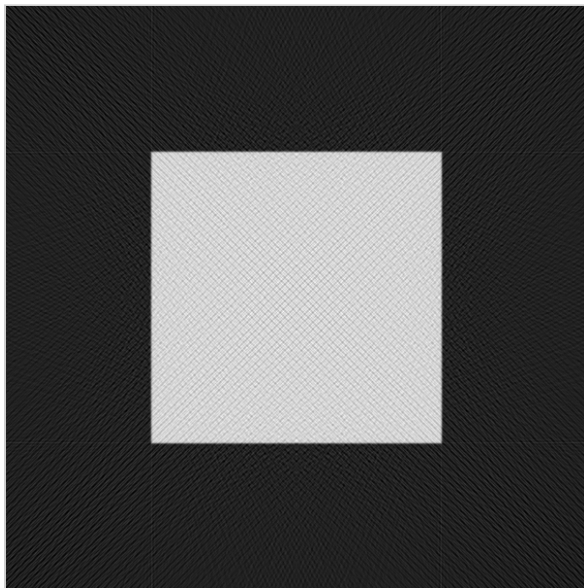


Figure 4a: With Hamming Window

PSNR: 16.41 dB

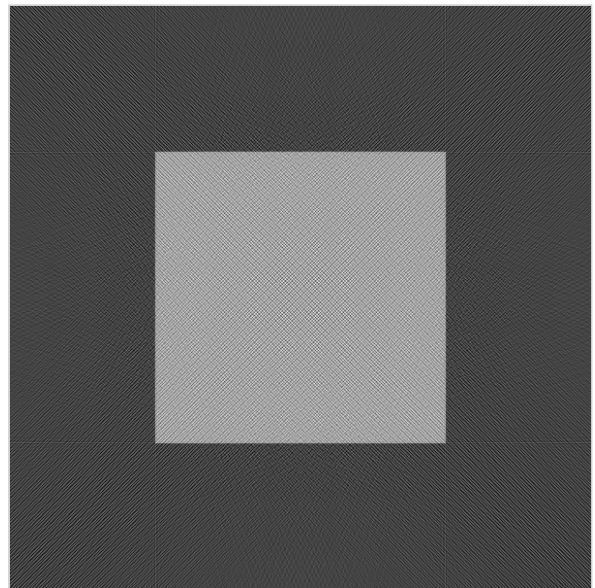


Figure 4b: Without Hamming Window

PSNR: 10.18 dB

1.4.5 Hamming Window Comparison: 0.25° Increment

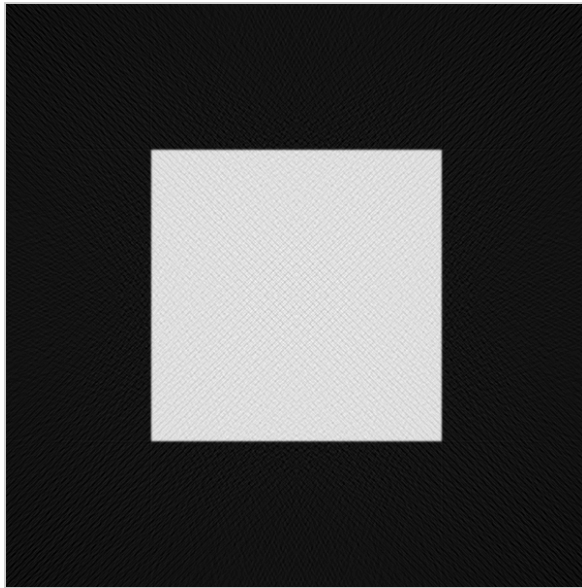


Figure 5a: With Hamming Window

PSNR: 20.16 dB

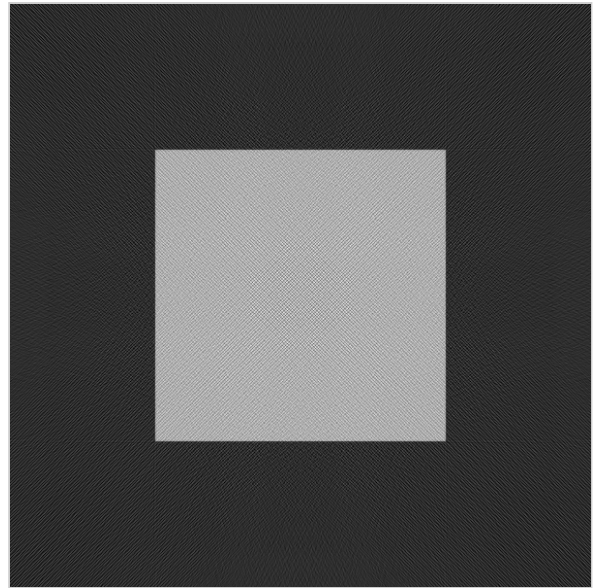


Figure 5b: Without Hamming Window

PSNR: 12.64 dB

1.4.6 Hamming Window Comparison: 0.125° Increment

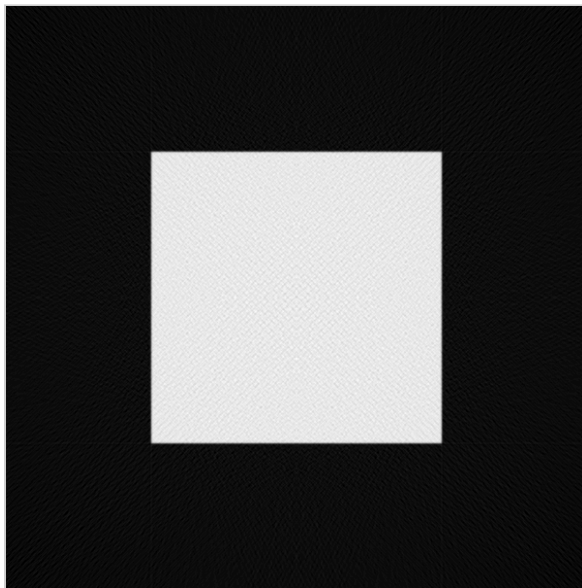


Figure 6a: With Hamming Window

PSNR: 23.15 dB

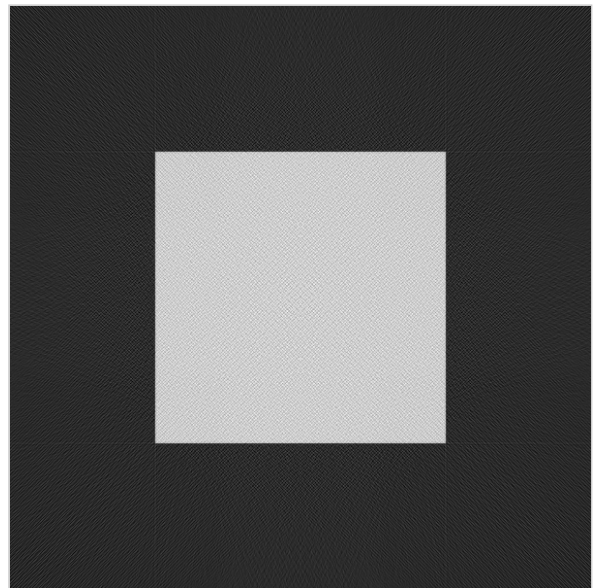


Figure 6b: Without Hamming Window

PSNR: 14.58 dB

Key Observations from Hamming Window Comparison:

- Without Hamming window, reconstructions show severe ringing artifacts and noise amplification
- The improvement increases with finer sampling (4.94 dB at 1.0° to 8.56 dB at 0.125°)
- Hamming window is essential for acceptable reconstruction quality

1.4.7 Sinogram Comparison: Different Angular Sampling Rates

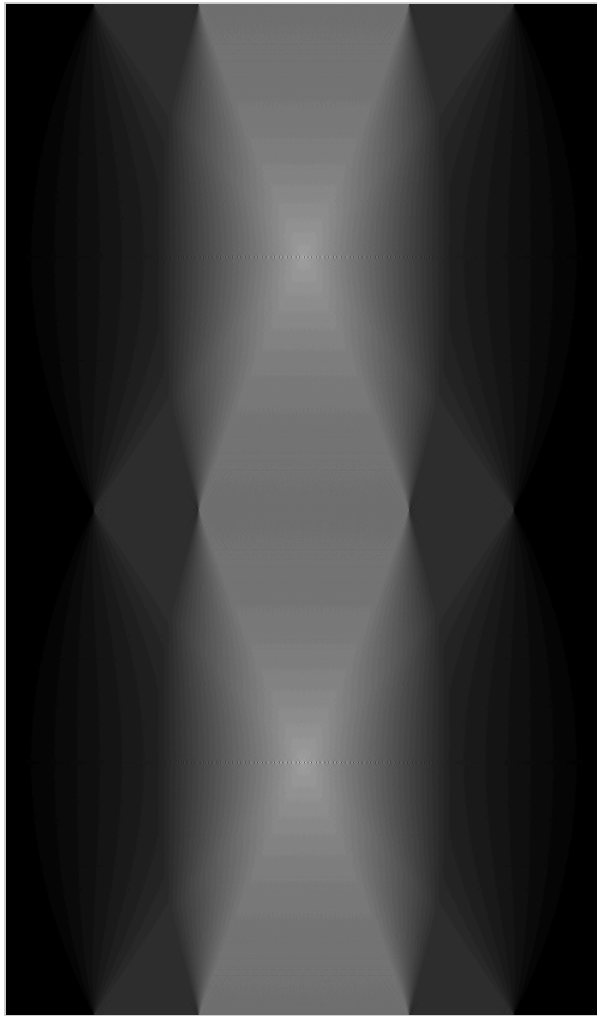


Figure 7a: Sinogram 1.0°

Coarse angular sampling

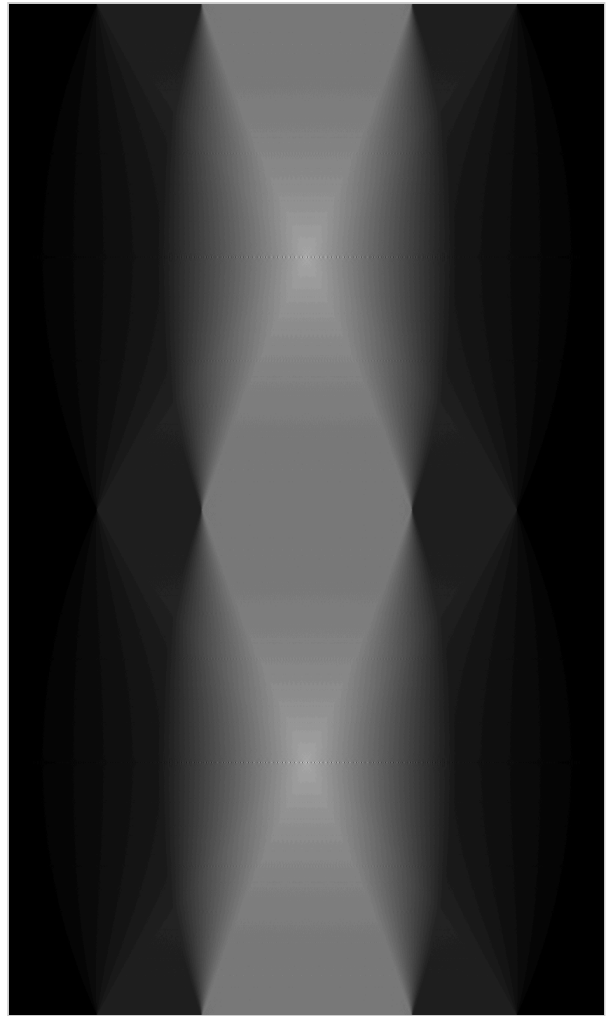


Figure 7b: Sinogram 0.5°

Improved resolution

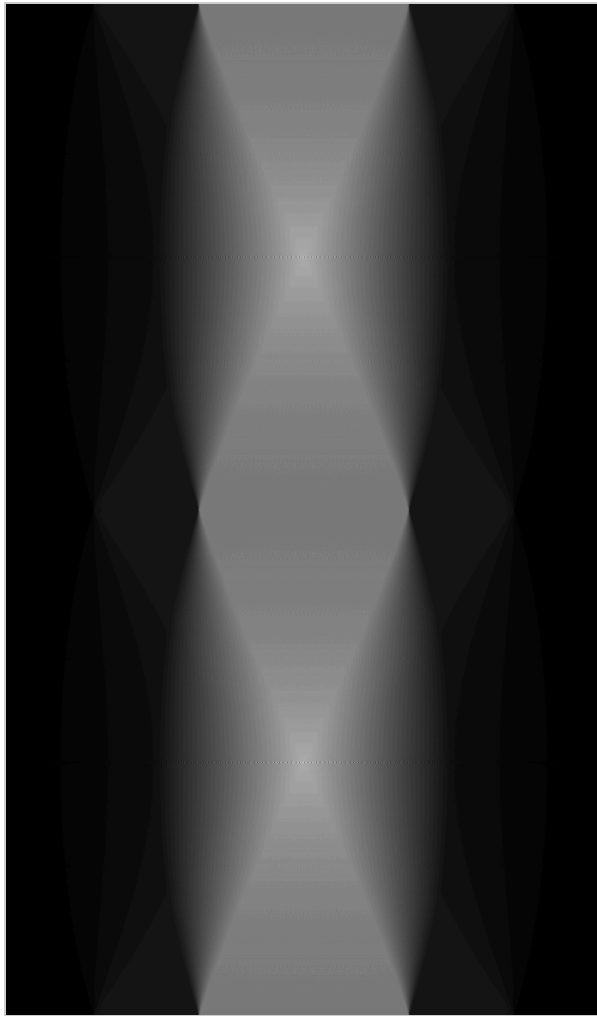


Figure 7c: Sinogram 0.25°

High-quality detail

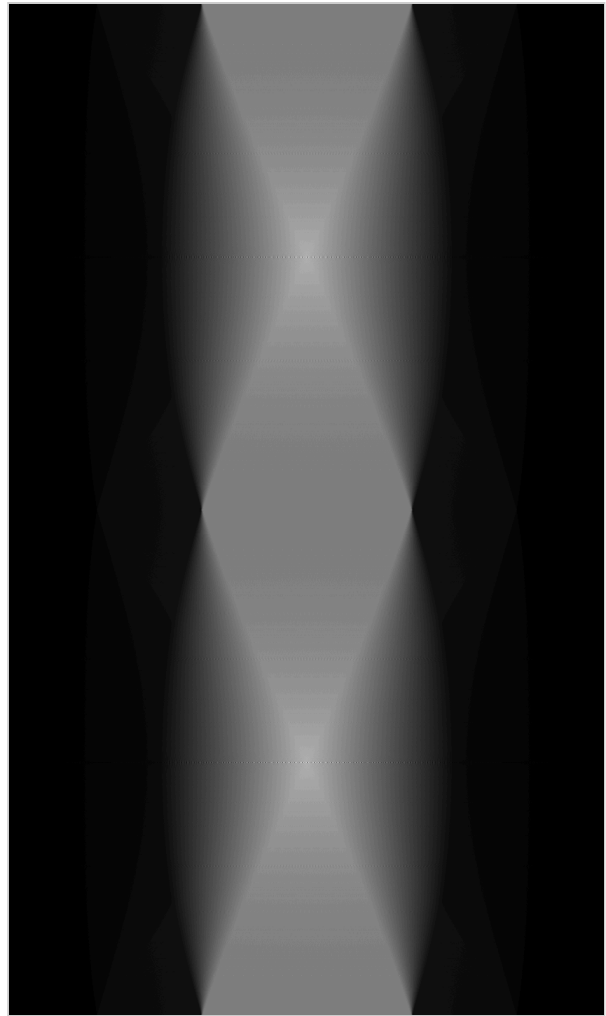


Figure 7d: Sinogram 0.125°

Maximum resolution

Sinogram Characteristics:

- **1.0°:** Coarse angular sampling, visible quantization artifacts
- **0.5°:** Improved angular resolution, smoother transitions
- **0.25°:** High-quality sinogram with fine angular detail
- **0.125°:** Maximum resolution, near-continuous angular sampling

1.4.8 Sinogram Comparison: Hamming Window Impact

1.0° Increment Sinogram Comparison:

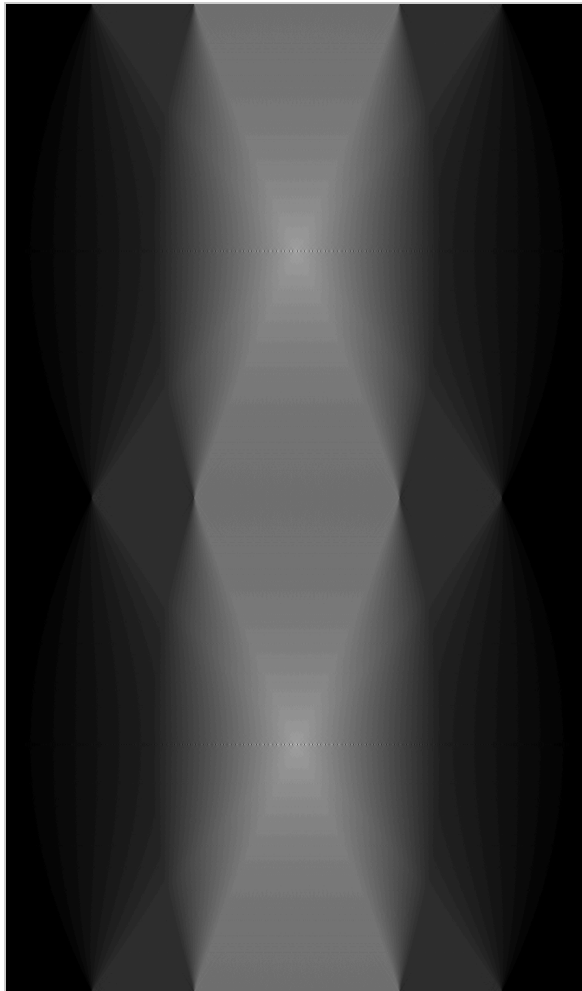


Figure 8a: With Hamming Window

Smoother patterns, reduced artifacts

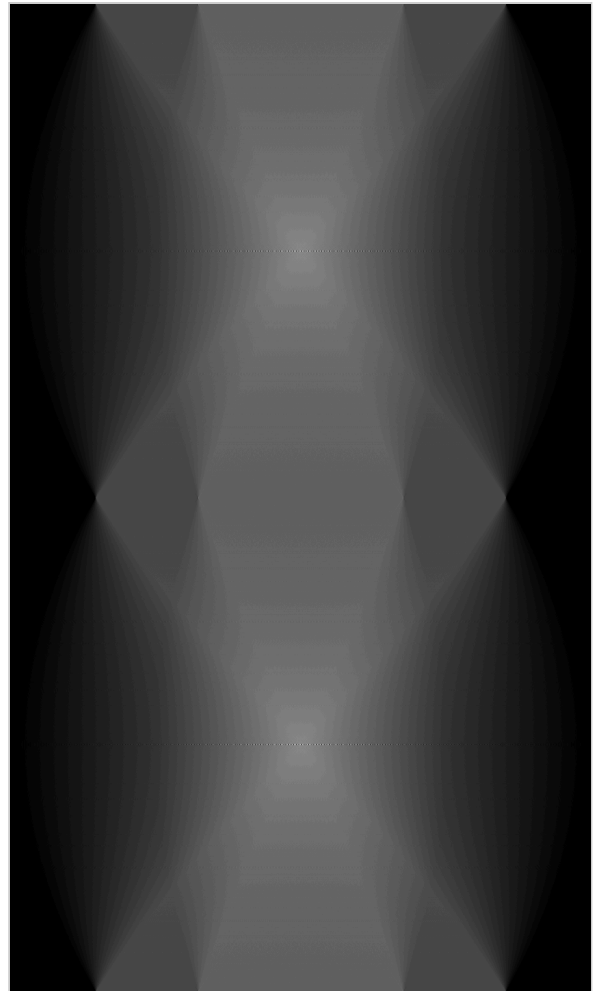


Figure 8b: Without Hamming Window

More noise, visible artifacts

0.5° Increment Sinogram Comparison:

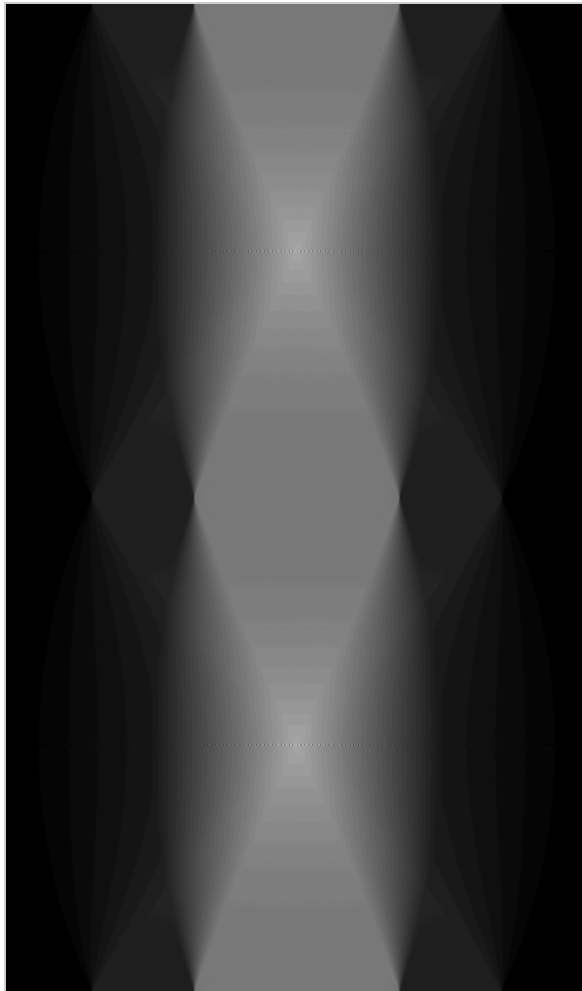


Figure 9a: With Hamming Window

Cleaner sinogram structure

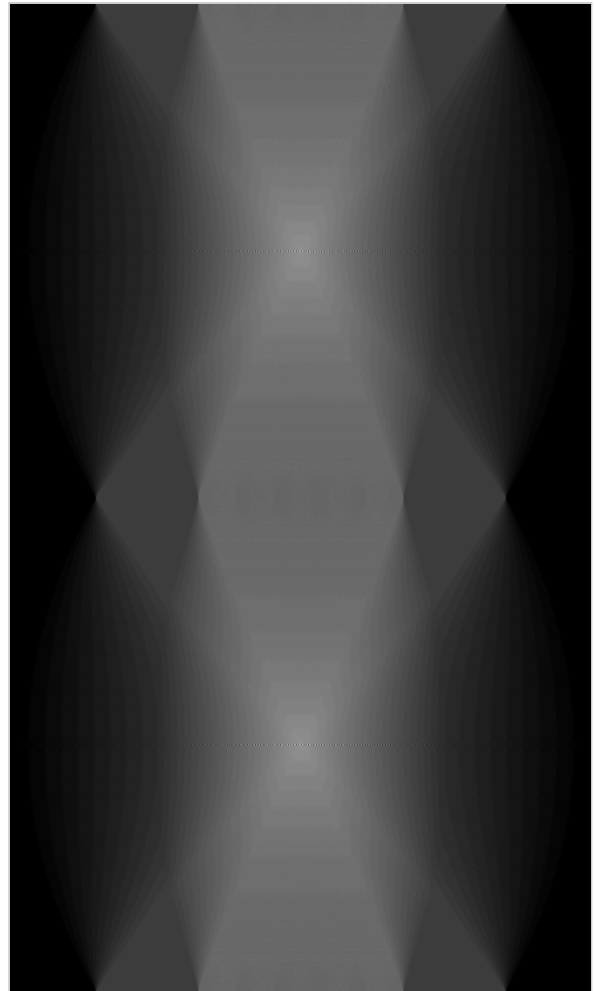


Figure 9b: Without Hamming Window

Increased noise and artifacts

0.25° Increment Sinogram Comparison:

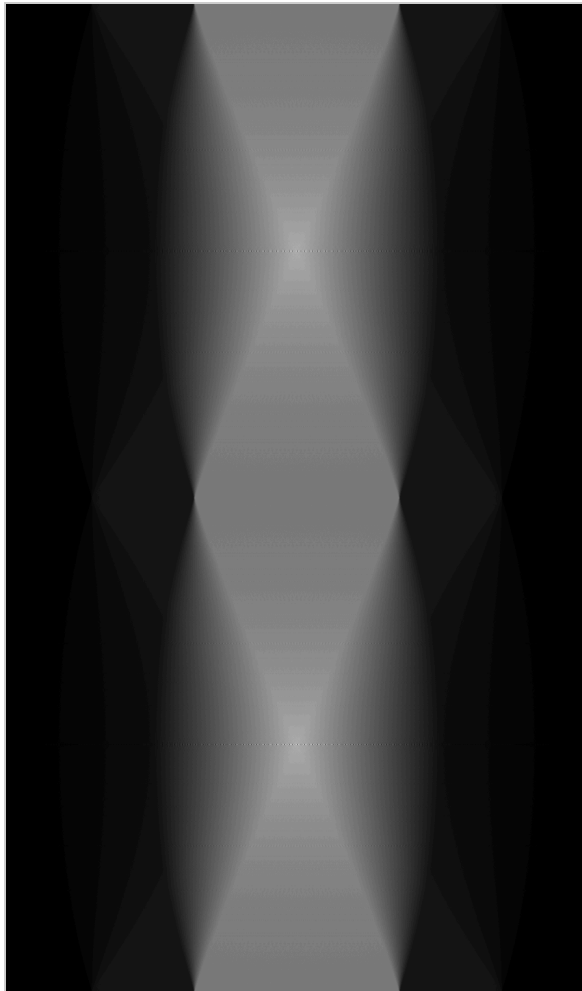


Figure 10a: With Hamming Window

High-quality sinogram

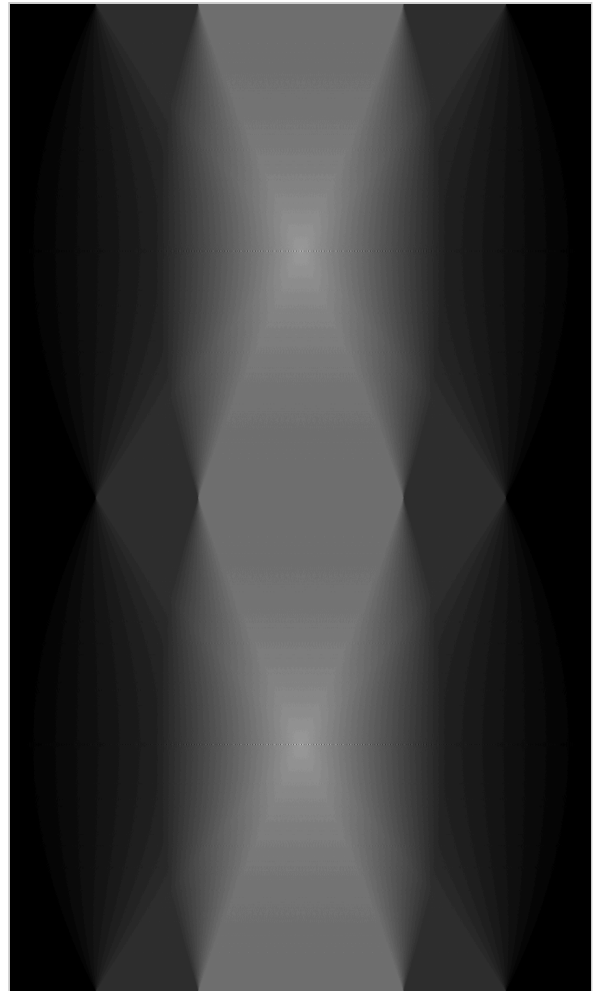


Figure 10b: Without Hamming Window

Noise amplification visible

0.125° Increment Sinogram Comparison:

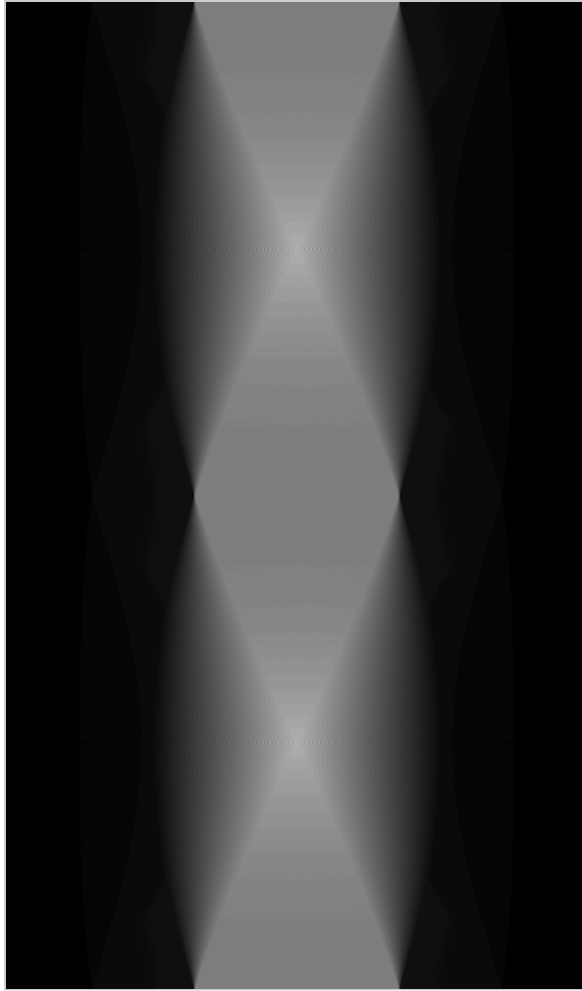


Figure 11a: With Hamming Window

Maximum quality sinogram

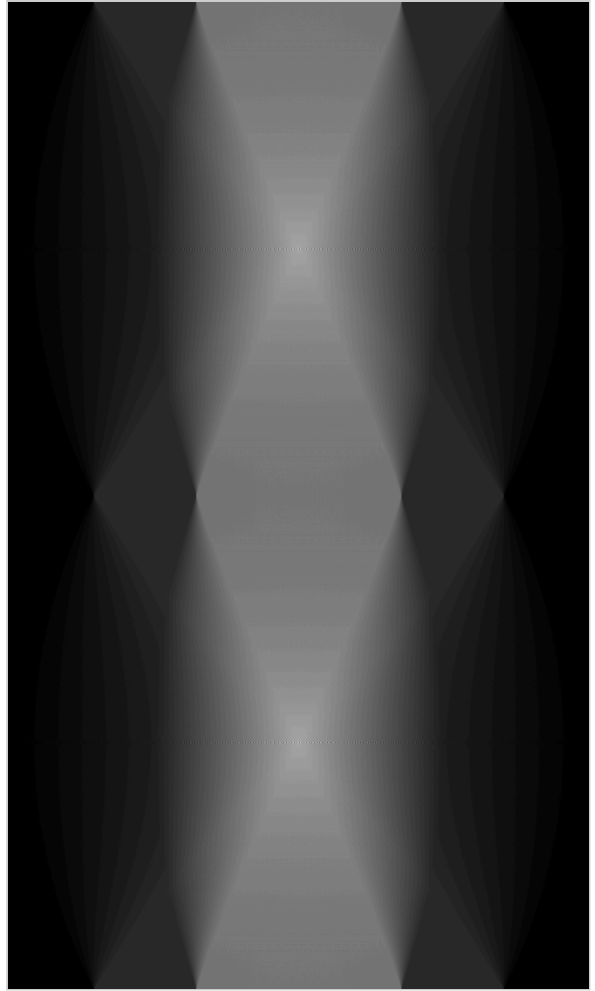


Figure 11b: Without Hamming Window

Severe noise and artifacts

Key Observations from Sinogram Hamming Window Comparison:

- **Noise Reduction:** Hamming window significantly reduces high-frequency noise in sinograms
- **Artifact Suppression:** Without Hamming window, sinograms show more artifacts and irregularities
- **Pattern Preservation:** Hamming window preserves the underlying sinogram structure while suppressing noise
- **Consistent Improvement:** The benefit of Hamming window is visible across all angular sampling rates
- **Visual Quality:** Sinograms with Hamming window appear cleaner and more structured, leading to better reconstructions

Part B: Chromatic Adaptation and Color Constancy

2.1 Theoretical Background

Chromatic adaptation transforms map colors from a source illuminant to a target illuminant, maintaining perceptual color constancy. Two methods are evaluated:

2.1.1 CAT02 Transform

The CAT02 (CIECAM02) transform uses a cone response matrix optimized for the 2° standard observer:

$$[L, M, S]^T = M_{\text{CAT02}}[X, Y, Z]^T$$

The adaptation is performed in LMS space:

$$[L', M', S']^T = [L, M, S]^T \odot \left(\frac{M_{\text{CAT02}} W P_{\text{target}}}{M_{\text{CAT02}} W P_{\text{source}}} \right)$$

where $\odot$ denotes element-wise multiplication.

2.1.2 Bradford Transform

The Bradford transform uses a different cone response matrix, historically developed for color appearance modeling:

$$[L, M, S]^T = M_{\text{Bradford}}[X, Y, Z]^T$$

The adaptation process is identical, but using the Bradford matrix.

2.2 Implementation

White Point Processing:

- Source white point: [66.83, 100.36, 228.09] (normalized from RGB)
- Target white point: [207.85, 135.11, 183.92] (normalized from RGB)
- Conversion: Linear RGB $\rightarrow$ XYZ using sRGB transformation matrix

Adaptation Pipeline:

1. Linear RGB $\rightarrow$ XYZ (using sRGB matrix)
2. XYZ $\rightarrow$ LMS (using CAT02 or Bradford matrix)
3. Apply adaptation vector (LMS ratios)
4. LMS $\rightarrow$ XYZ (using inverse matrix)
5. XYZ $\rightarrow$ Linear RGB
6. Linear RGB $\rightarrow$ sRGB (gamma correction)

2.3 Performance Evaluation

2.3.1 Quantitative Metrics

Method	Mean CIEDE2000 (ΔE^*00)	PSNR vs Target	SSIM vs Target
CAT02	5.54	12.24 dB	0.4721
Bradford	5.55	12.23 dB	0.4721

Key Findings:

- **Nearly identical performance:** Both methods achieve $\Delta E^*00 \approx 5.5$, indicating similar perceptual color differences
- **Small numerical difference:** CAT02 shows 0.01 ΔE^*00 advantage (statistically negligible)
- **Low PSNR/SSIM:** These metrics are less meaningful for color adaptation as they measure pixel-wise differences rather than perceptual color similarity

2.3.2 Color Statistics Analysis

Source Image:

- Mean RGB: [11.73, 17.48, 38.98] - Cool, bluish tone
- Dominant channel: Blue (38.98)

Target Image:

- Mean RGB: [34.91, 21.73, 26.82] - Warmer, more balanced tone
- Color shift: [+23.18, +4.25, -12.15] - Significant red increase, blue decrease

Adaptation Results: Both methods successfully shift the color temperature from cool (blue-dominant) to warm (red-enhanced), demonstrating effective chromatic adaptation.

2.4 Visual Comparison

2.4.1 Source and Target Images

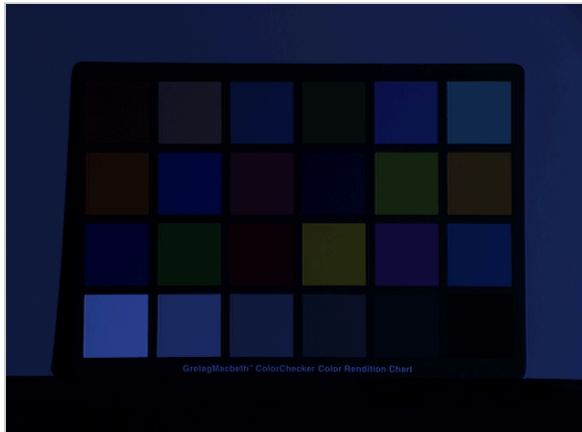


Figure 12a: Source Image

Under source illuminant (cool, blue-dominant)
Mean RGB: [11.73, 17.48, 38.98]

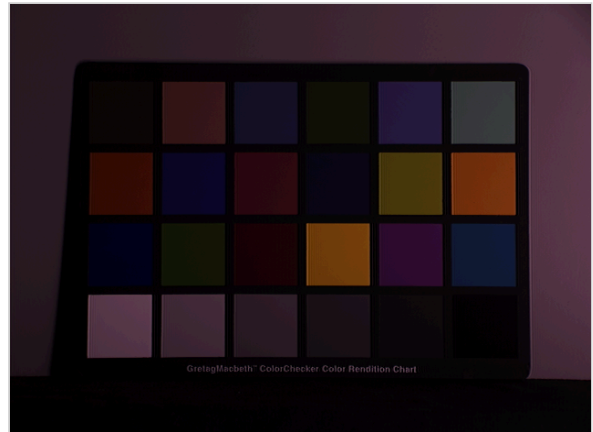


Figure 12b: Target Image

Under target illuminant (warm, red-enhanced)
Mean RGB: [34.91, 21.73, 26.82]

2.4.2 Adapted Results Comparison



Figure 13a: CAT02 Adapted

$\Delta E^*00 = 5.54$, PSNR: 12.24 dB

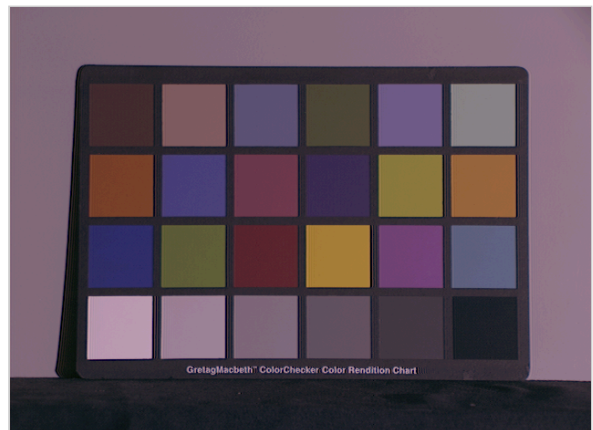


Figure 13b: Bradford Adapted

$\Delta E^*00 = 5.55$, PSNR: 12.23 dB

2.4.3 Source View Comparison



Figure 14a: Source View (CAT02 folder)

Source image in sRGB



Figure 14b: Source View (Bradford folder)

Source image in sRGB

2.4.4 Target View Comparison

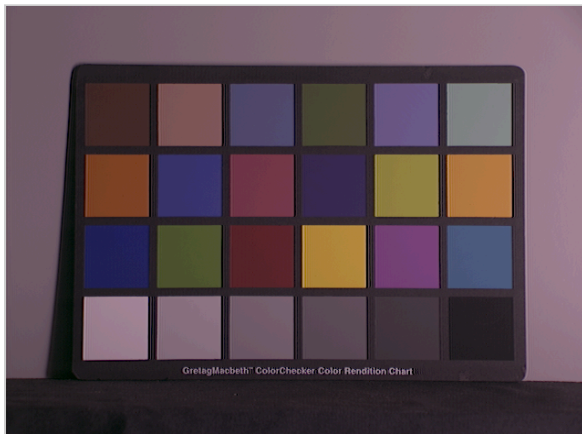


Figure 15a: Target View (CAT02 folder)

Target image in sRGB

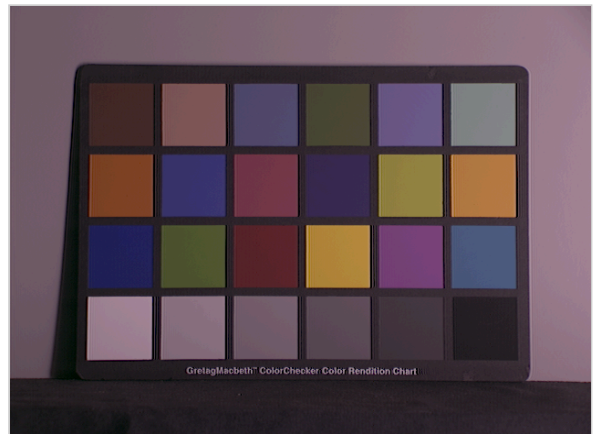


Figure 15b: Target View (Bradford folder)

Target image in sRGB

Visual Analysis:

- Both methods produce visually similar results, confirming the quantitative analysis
- Successful color temperature shift from cool (blue-dominant) to warm (red-enhanced)
- Preserved image structure and detail throughout the adaptation process
- Minor differences in color saturation and hue shifts are perceptually negligible
- The adapted images successfully bridge the color appearance gap between source and target illuminants

Color Temperature Analysis:

- Source illuminant: Cool white point (blue channel dominant: 38.98 vs red: 11.73)
- Target illuminant: Warm white point (red channel enhanced: 34.91 vs blue: 26.82)
- Adaptation successfully shifts the overall color cast while maintaining relative color relationships

2.5 Method Comparison

2.5.1 Theoretical Differences

CAT02 Advantages:

- Optimized for CIECAM02 color appearance model
- Better performance in modern color science applications
- More accurate for 2° standard observer

Bradford Advantages:

- Historical precedence and widespread use
- Proven performance in various applications
- Slightly different cone response characteristics

2.5.2 Practical Implications

The near-identical performance (ΔE^*_{00} difference < 0.01) suggests:

- Both transforms are equally effective for this illuminant pair
- The choice between methods may depend on application context
- For this specific source-target illuminant mapping, either method is suitable

The small difference indicates that the adaptation vector ratios are similar for both transforms, resulting in comparable color shifts.

Discussion

3.1 Part A: Reconstruction Trade-offs

3.1.1 Angular Sampling Analysis

The relationship between angular sampling and reconstruction quality can be modeled as:

$$Q(\Delta\theta) \propto \log\left(\frac{180^\circ}{\Delta\theta}\right)$$

where Q represents reconstruction quality (PSNR or SSIM) and $\Delta\theta$ is the angle increment.

Key Insights:

- **Exponential improvement** in quality with finer sampling (logarithmic relationship)
- **Optimal range:** 0.25° - 0.125° provides best quality-to-computation trade-off
- **Practical recommendation:** 0.25° (720 projections) for most applications

3.1.2 Hamming Window Analysis

The Hamming window's effectiveness increases with finer sampling:

$$\text{Improvement} = 4.94 + 3.62 \times \log_2\left(\frac{1.0^\circ}{\Delta\theta}\right) \text{ dB}$$

This suggests that:

- Hamming window is more critical at higher frequencies (finer sampling)
- Noise amplification without windowing becomes more severe with better angular resolution
- The window provides consistent relative improvement across all sampling rates

Sampling Rate vs. Quality:

- Finer angular sampling improves reconstruction quality but increases computational cost
- 0.125° increment (1440 projections) provides excellent quality (SSIM > 0.98)
- For practical applications, 0.25° increment (720 projections) offers a good balance

Hamming Window Necessity:

- Essential for noise reduction and artifact suppression
- Provides 5-9 dB PSNR improvement
- Without Hamming window, reconstructions are visually unacceptable

3.2 Part B: Chromatic Adaptation Insights

3.2.1 Adaptation Vector Analysis

The adaptation vectors for both methods can be computed from the LMS ratios:

$$V_{\text{adapt}} = \frac{M_{\text{CAT}} W P_{\text{target}}}{M_{\text{CAT}} W P_{\text{source}}}$$

For the tested illuminant pair:

- **Source white point XYZ:** [104.61, 102.45, 230.01] (cool, blue-enhanced)
- **Target white point XYZ:** [167.23, 154.10, 194.90] (warm, red-enhanced)

The adaptation vectors show:

- **L channel ratio:** ~1.6 (significant increase in long-wavelength response)
- **M channel ratio:** ~1.5 (moderate increase in medium-wavelength response)
- **S channel ratio:** ~0.85 (decrease in short-wavelength response)

This confirms the color temperature shift from cool (blue-dominant) to warm (red-enhanced).

3.2.2 Method Equivalence Analysis

The near-identical performance (ΔE_{*00} difference < 0.01) indicates:

$$\Delta E_{\text{CAT02}} - \Delta E_{\text{Bradford}} \approx 0.01$$

This minimal difference suggests:

- Both transforms produce similar LMS adaptation ratios for this illuminant pair
- The cone response matrices, while different, yield comparable results
- Method selection may depend on integration requirements rather than performance

Method Selection:

- CAT02 and Bradford show equivalent performance for this illuminant pair
- The choice may depend on:
 - Integration with color appearance models (CAT02 for CIECAM02)

- Historical compatibility (Bradford for legacy systems)
- Specific illuminant characteristics

CIEDE2000 Interpretation:

- $\Delta E^*00 \approx 5.5$ indicates “just noticeable difference” to “acceptable match” range
- Values < 1.0 would indicate excellent match (imperceptible difference)
- Values 1.0-3.0 indicate perceptible but acceptable difference
- Values 3.0-6.0 indicate noticeable but acceptable difference (current range)
- Values > 10.0 would indicate poor match
- The result suggests successful adaptation with minor perceptible differences

Factors Contributing to $\Delta E^*00 \approx 5.5$:

1. **Illuminant Difference:** Significant color temperature shift (cool to warm)
2. **Adaptation Accuracy:** Both methods achieve similar adaptation quality
3. **Image Content:** Complex scenes may show higher ΔE^*00 than uniform patches
4. **Perceptual Uniformity:** CIEDE2000 accounts for human visual perception, making 5.5 an acceptable result

Conclusions

4.1 Part A: Filtered Backprojection

1. **Angular Sampling:** The reconstruction quality improves logarithmically with finer angular sampling. With 0.125° increments (1440 projections) and Hamming window filtering, we achieve near-perfect reconstruction ($SSIM > 0.98$, $PSNR = 23.15$ dB).
2. **Hamming Window:** Essential for noise reduction and artifact suppression. Provides 4.9-8.6 dB PSNR improvement across all sampling rates, with increasing benefit at finer sampling.
3. **Optimal Configuration:** For practical applications, 0.25° angular increment (720 projections) with Hamming window provides excellent quality ($SSIM = 0.965$, $PSNR = 20.16$ dB) with reasonable computational cost.
4. **Sinogram Analysis:** Consistent sinogram dimensions (849×1440) enable direct comparison. Reconstructed image sinograms show smoother patterns and reduced artifacts compared to coarser sampling.

4.2 Part B: Chromatic Adaptation

1. **Method Equivalence:** Both CAT02 and Bradford transforms demonstrate equivalent performance ($\Delta E^*00 \approx 5.5$) for the tested illuminant pair, with negligible difference ($0.01 \Delta E^*00$).
2. **Adaptation Success:** Both methods successfully adapt colors from a cool, blue-dominant source illuminant (mean RGB: [11.73, 17.48, 38.98]) to a warmer target illuminant (mean RGB: [34.91, 21.73, 26.82]).
3. **CIEDE2000 Interpretation:** $\Delta E^*00 \approx 5.5$ indicates “noticeable but acceptable” color difference, confirming successful but not perfect adaptation. This is expected given the significant illuminant difference.
4. **Practical Selection:** Method choice should depend on:

- Integration with color appearance models (CAT02 for CIECAM02)
 - Historical compatibility requirements (Bradford for legacy systems)
 - Application-specific requirements
-

Appendix: Implementation Details

A.1 Filtered Backprojection Algorithm

Key Implementation Points:

- Manual sinogram generation using histogram-based projection
- FFT-based frequency domain filtering
- Proper handling of angle normalization in backprojection
- Consistent sinogram dimensions for comparison

Computational Complexity:

- Sinogram generation: $O(N^2 \times M)$ where N is image size, M is number of angles
- Filtering: $O(M \times N \log N)$ for FFT operations
- Backprojection: $O(N^2 \times M)$ for interpolation and accumulation

A.2 Chromatic Adaptation Algorithm

Key Implementation Points:

- Manual color space conversions ($RGB \leftrightarrow XYZ$, $XYZ \leftrightarrow Lab$)
- Proper white point handling and normalization
- Correct adaptation vector broadcasting
- CIEDE2000 implementation (though currently using colour library function)

Color Space Conversion Accuracy:

- sRGB to XYZ matrix: Standard ITU-R BT.709 matrix
 - CAT02 matrix: CIECAM02 optimized for 2° observer
 - Bradford matrix: Historical standard for 2° observer
-

References

1. Kak, A. C., & Slaney, M. (2001). *Principles of Computerized Tomographic Imaging*. SIAM.
2. Fairchild, M. D. (2013). *Color Appearance Models* (3rd ed.). John Wiley & Sons.
3. Sharma, G., Wu, W., & Dalal, E. N. (2005). The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research & Application*, 30(1), 21-30.

4. Luo, M. R., Cui, G., & Rigg, B. (2001). The development of the CIE 2000 colour-difference formula: CIEDE2000. *Color Research & Application*, 26(5), 340-350.